

Venkata Sainadh Mannam

IAM Engineer

SUMMARY

IAM Engineer with 6+ years of experience designing and securing large-scale identity ecosystems across cloud and enterprise environments. Skilled in modern access management, authentication, and governance frameworks with hands-on expertise in Okta, Azure AD, AWS IAM, CyberArk, and Identity Threat Detection and Response. Strong background in automating identity lifecycle workflows, architecting SSO and MFA solutions, and building least-privilege access models. Experienced in applying machine learning for risk-based authentication, behavioral analytics, and identity risk scoring. Known for improving security posture, reducing access friction, and aligning complex IAM programs with Zero Trust and key compliance standards including SOX, HIPAA, PCI-DSS, and SOC 2.

TECHNICAL SKILLS

Identity & Access Management: Okta (UD, SAML, OIDC, Workflows), Azure AD / Entra ID (Conditional Access, App Registrations, PIM), SailPoint (IIQ, IDN), CyberArk (PAM), ADFS, LDAP, SCIM provisioning, MFA integration.

Security & Governance: RBAC, ABAC, identity lifecycle automation, Zero Trust, IGA policies, SoD controls, certificate and key management, access reviews, compliance alignment (SOX, HIPAA, PCI-DSS, SOC 2).

Cloud & Infrastructure: AWS IAM & Cognito, Azure IAM, Active Directory, Azure AD Connect, directory services, federation architecture, API security.

Automation & Scripting: Python, PowerShell, Bash, REST APIs, Terraform (IAM automation), GitHub Actions, CI/CD workflows, Okta Workflows, SailPoint Rules.

ML & AI for Identity Security: risk-based authentication models, behavioral biometrics analysis (login time, geo-location), anomaly detection, user-pattern clustering, ML-powered access insights, AI-driven threat detection, ITDR signal correlation.

Authentication Protocols: SAML 2.0, OAuth 2.0, OIDC, Kerberos, JWT, SCIM, WS-Fed.

PROFESSIONAL EXPERIENCE

CrowdStrike | TX | IAM Engineer (AI & ML)

August 2024 – Present

- Architected and deployed a **Risk-Based Authentication (RBA)** engine by integrating **ML models** into the **CrowdStrike Falcon** platform, enabling real-time contextual access decisions and reducing authentication friction for **15,000+ users**.
- Engineered dynamic access policies using **behavioral biometrics**, **geo-location**, and **device posture analytics** to trigger **adaptive MFA**, increasing risky login detection accuracy by **30%**.
- Integrated **CyberArk Privileged Access Management (PAM)** with **AI/ML risk scoring pipelines** to assess service account behavior and reduce exposure windows by **50%**.
- Developed and optimized **Python** and **PowerShell** scripts for automated **Joiner-Mover-Leaver (JML)** provisioning workflows across **Okta** and **Azure AD**, achieving **90% automation** and saving **20 hours per week** in manual effort.
- Designed and implemented **Identity Threat Detection and Response (ITDR)** alerting using **Splunk** and **CrowdStrike Identity Protection logs**, containing compromised identities with an average **7-minute MTTR**.
- Automated **Identity Governance** and **Administration (IGA)** reporting and compliance workflows aligned with **SOX** and **HIPAA**, centralizing access attestations and cutting **audit preparation time** by **12%**.

Okta | CA | IAM Security Engineer

January 2023 – August 2024

- Designed and implemented **Single Sign-On (SSO)** architecture using **SAML 2.0**, **OAuth 2.0**, and **OpenID Connect (OIDC)** protocols to integrate key **SaaS apps** into **Okta Universal Directory**, reducing password-related support tickets by **30%**.
- Led the end-to-end migration of **15,000+ users** and **150 legacy applications** from **Active Directory Federation Services (AD FS)** and **on-prem LDAP** to the **Okta Workforce Identity Cloud (WIC)**, achieving a **95% user adoption rate** post-cutover.
- Developed advanced **adaptive security policies** leveraging **Okta Adaptive MFA** and **Okta ThreatInsight**, automatically blocking **3,000+ high-risk sign-on attempts monthly** using **IP reputation**, **geo-velocity**, and **device trust analytics**.
- Automated **Joiner-Mover-Leaver (JML)** lifecycle management via **Okta Lifecycle Management (LCM)** and **SCIM integrations** with **Azure AD** and **Workday**, cutting **account creation time** from hours to under **30 minutes**.
- Created custom **API hooks** and **inline federation scripts** using **Okta Expression Language (EL)** to enforce **granular RBAC** and **identity transformation logic**, ensuring compliant access control for privileged and sensitive roles.
- Partnered with **enterprise security teams** to align Okta configurations with **PCI-DSS** and **SOC 2 compliance frameworks**, successfully passing **two external audits** and strengthening the organization's **Zero Trust security posture**.

Amazon | India | IAM Engineer

March 2018 – July 2021

- Designed a **multi-account AWS IAM architecture** across **40+ production** and **development accounts** using **AWS Organizations** and **Service Control Policies (SCPs)**, standardizing global security controls and enforcing mandatory guardrails.
- Developed and maintained **least-privilege IAM policies** in **JSON** for **8,000+ AWS principals** (users, roles, and groups), achieving a **15% reduction in over-privileged identities** and minimizing the **overall attack surface**.
- Engineered a **cross-account role assumption framework** leveraging **IAM roles**, **external IDs**, and **CI/CD tools** such as **Jenkins** and **AWS CodePipeline**, strengthening **deployment security** across environments.
- Implemented **Multi-Factor Authentication (MFA)** for all privileged access and introduced **Conditional Access policies** based on **source IP** and **time-of-day**, enhancing **AWS environment** resilience against unauthorized access.
- Automated detection and remediation of **IAM policy non-compliance** (wildcard permissions, inactive keys, root account usage) through **AWS Config Rules** and **Lambda (Python)**, reducing **policy violation resolution time** from days to hours.
- Conducted periodic **access reviews** and **entitlement audits** using **AWS Access Analyzer** and **CloudTrail**, embedding **Identity Governance and Administration (IGA)** best practices and ensuring compliance with internal and external security standards.

EDUCATION

Master's in Information Technology | University of Cumberlands, Kentucky, USA.

CERTIFICATES

- **Okta Professional and Okta Administrator**